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PRN: 24070521062
Experiment / Case Study No.: 10

1. Experiment Title:

Load and Display JSON Data in a Web Page Using the Fetch API (Part 1) and jQuery `$.getJSON()` (Part 2)

2. Software/Tools Required:

Visual Studio Code, HTML5, CSS3, JavaScript (Fetch API), jQuery 3.7.1, JSON, Google Chrome Browser / Microsoft Edge

3. Theory:

Part 1 — Fetch API. JSON (JavaScript Object Notation) is a lightweight text format used to exchange data between a server and a web page. The `fetch()` method is the modern, promise-based way of requesting such a resource from JavaScript. It takes the URL of the file, returns a Promise that resolves to a `Response` object, and the method `response.json()` then parses the body into a JavaScript array or object. The result is handled in a `.then()` chain, in which the rows of an HTML table are built dynamically, and any network or parsing failure is caught in `.catch()`. The page is never reloaded because the request is asynchronous.

Part 2 — jQuery. jQuery is a JavaScript library that shortens DOM selection, event handling and AJAX calls. Its helper `$.getJSON(url, callback)` is a shorthand for `$.ajax()` with the data type set to JSON: the file is requested asynchronously and the parsed data is passed to the success callback. `$.each()` iterates over the records, `.append()` adds the rows to the table, `.empty()` clears the old rows and `.fail()` handles an error. Compared with Fetch, the same task needs less code because jQuery handles the response parsing internally.

4. Experiment Program Code:

Part 1 — Fetch API

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Load JSON using Fetch</title>
  <style>
    table {
      width: 70%;
      border-collapse: collapse;
      margin-top: 20px;
    }
    th, td {
      border: 1px solid black;
      padding: 10px;
      text-align: left;
    }
    th {
      background-color: #f2f2f2;
    }
    button {
      padding: 10px 20px;
      font-size: 16px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h2>Student Details using Fetch()</h2>
```

```
  <button onclick="loadStudents()">Load Students</button>

  <table id="studentTable">
    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Marks</th>
      </tr>
    </thead>
    <tbody id="tableBody">
    </tbody>
  </table>

  <script>
    function loadStudents() {
      fetch("students.json")
        .then(function(response) {
          return response.json();
        })
        .then(function(data) {
          var tbody = document.getElementById("tableBody");
          tbody.innerHTML = "";
          data.forEach(function(student) {
            var row = "<tr>" +
              "<td>" + student.id + "</td>" +
              "<td>" + student.name + "</td>" +
              "<td>" + student.department + "</td>" +
              "<td>" + student.marks + "</td>" +
              "</tr>";
            tbody.innerHTML += row;
          });
        })
        .catch(function(error) {
          alert("Failed to fetch: " + error.message);
        });
    }
  </script>
</body>
</html>
```

students.json

```
[
  { "id": 1, "name": "Aarav Sharma", "department": "Computer Science", "marks": 92 },
  { "id": 2, "name": "Priya Patel", "department": "Information Technology", "marks": 88 },
  { "id": 3, "name": "Rohan Mehta", "department": "Electronics", "marks": 75 },
  { "id": 4, "name": "Sneha Joshi", "department": "Computer Science", "marks": 95 },
  { "id": 5, "name": "Karan Desai", "department": "Mechanical", "marks": 70 }
]
```

Part 2 — jQuery

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>JSON using jQuery</title>
  <!-- jQuery Library -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
  <style>
    table {
      border-collapse: collapse;
      width: 70%;
      margin-top: 20px;
    }
    th, td {
      border: 1px solid black;
      padding: 10px;
      text-align: center;
    }
    th {
      background-color: lightgray;
    }
    button {
```

```
      padding: 10px 20px;
      font-size: 16px;
    }
  </style>
</head>
<body>
  <h2>Student Details</h2>
  <button id="loadBtn">Load Students</button>
  <table id="studentTable">
    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Marks</th>
      </tr>
    </thead>
    <tbody id="tableBody">
    </tbody>
  </table>

  <script>
    $("#loadBtn").click(function() {
      $.getJSON("students.json", function(data) {
        $("#tableBody").empty();
        $.each(data, function(index, student) {
          $("#tableBody").append(`
            <tr>
              <td>${student.id}</td>
              <td>${student.name}</td>
              <td>${student.department}</td>
              <td>${student.marks}</td>
            </tr>
          `);
        });
      }).fail(function() {
        alert("Error loading JSON data.");
      });
    });
  </script>
</body>
</html>
```

students.json (same data file as Part 1)

5. Output:

Student Details using Fetch()

Load Students

ID	Name	Department	Marks
1	Aarav Sharma	Computer Science	92
2	Priya Patel	Information Technology	88
3	Rohan Mehta	Electronics	75
4	Sneha Joshi	Computer Science	95
5	Karan Desai	Mechanical	70

Fig 1: Part 1 (Fetch API) output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 10/part 1/index.html

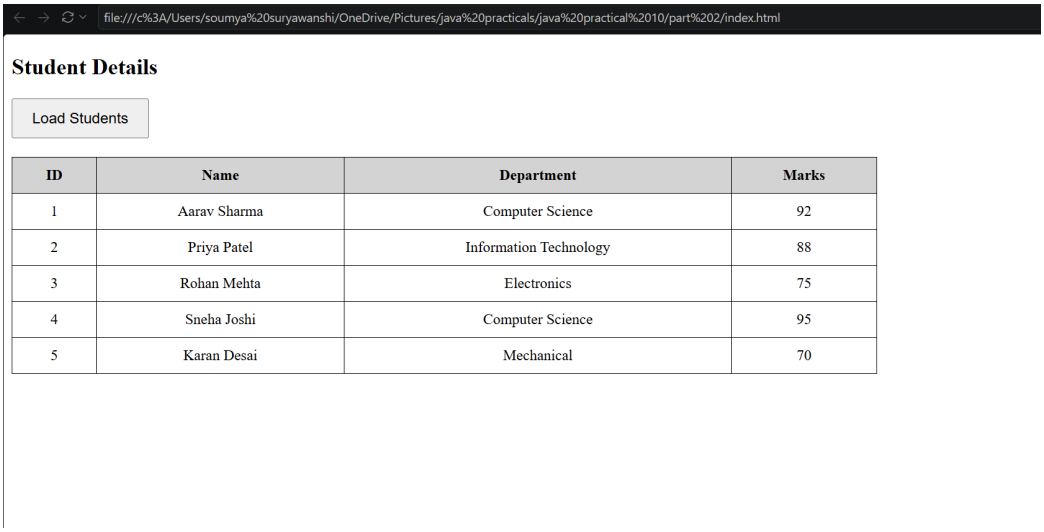


Fig 2: Part 2 (jQuery) output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 10/part 2/index.html

6. Case Study Title:

Part 1: Employee Manager — Adding Employee Records as JSON Objects and Loading `employees.json` with the Fetch API
Part 2: Online Food Order Dashboard — Loading `orders.json` with jQuery `$.getJSON()` and Calculating an Order Summary

7. Case Study Program Code:

Part 1 — Employee Manager (Fetch API)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Employee Manager</title>
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap" rel="styleshee
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <header>
            <h1><span class="icon">&#128101;</span> Employee Manager</h1>
            <p>Manage employee records using JavaScript &amp; JSON</p>
        </header>

        <div class="grid">
            <div class="card form-card">
                <h2>Add Employee</h2>
                <form id="empForm">
                    <div class="input-group">
                        <label for="empId">Employee ID</label>
                        <input type="text" id="empId" placeholder="e.g. 101" required>
                    </div>
                    <div class="input-group">
                        <label for="empName">Employee Name</label>
                        <input type="text" id="empName" placeholder="e.g. Alice Johnson" required>
                    </div>
                    <div class="input-group">
                        <label for="dept">Department</label>
                        <input type="text" id="dept" placeholder="e.g. Engineering" required>
                    </div>
                    <div class="input-group">
                        <label for="salary">Salary</label>
                        <input type="number" id="salary" placeholder="e.g. 75000" required>
```

```
                </div>
                <button type="submit" class="btn btn-green">&#10133; Add Employee</button>
            </form>
        </div>

        <div class="card table-card">
            <div class="table-header">
                <h2>Employee Records</h2>
                <button id="loadBtn" class="btn btn-blue">&#11015; Load Employees</button>
            </div>
            <table id="empTable">
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>Name</th>
                        <th>Department</th>
                        <th>Salary</th>
                    </tr>
                </thead>
                <tbody></tbody>
            </table>
            <p id="emptyMsg" class="empty-msg">No records yet — add one or load from JSON.</p>
            <div class="json-output">
                <div class="json-header">
                    <h3>Live JSON Output</h3>
                    <button id="saveBtn" class="btn btn-purple">&#128190; Save as JSON</button>
                </div>
                <pre id="jsonDisplay"></pre>
            </div>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

script.js

```
const employees = [];
const tbody = document.querySelector('#empTable tbody');
const emptyMsg = document.getElementById('emptyMsg');
const jsonDisplay = document.getElementById('jsonDisplay');

// Add Employee via form
document.getElementById('empForm').addEventListener('submit', function(e) {
    e.preventDefault();

    const empObj = {
        id: document.getElementById('empId').value,
        name: document.getElementById('empName').value,
        department: document.getElementById('dept').value,
        salary: document.getElementById('salary').value
    };

    JSON.stringify(empObj); // convert to JSON string

    employees.push(empObj);
    addRow(empObj);
    updateJsonDisplay();
    this.reset();
});

// Load employees from employees.json using Fetch API
document.getElementById('loadBtn').addEventListener('click', function() {
    fetch('employees.json')
        .then(function(response) {
            return response.json();
        })
        .then(function(data) {
            data.forEach(function(emp) {
                employees.push(emp);
                addRow(emp);
            });
            updateJsonDisplay();
        })
        .catch(function(error) {
            alert('Failed to load employees.json: ' + error.message);
```

```
        });
});

function addRow(emp) {
    emptyMsg.style.display = 'none';
    const row = document.createElement('tr');
    row.innerHTML = `
        <td>${emp.id}</td>
        <td>${emp.name}</td>
        <td>${emp.department}</td>
        <td>$${Number(emp.salary).toLocaleString()}</td>
    `;
    tbody.appendChild(row);
}

function updateJsonDisplay() {
    jsonDisplay.textContent = JSON.stringify(employees, null, 2);
}

// Save current employees array as JSON file
document.getElementById('saveBtn').addEventListener('click', function() {
    const blob = new Blob([JSON.stringify(employees, null, 2)], { type: 'application/json' });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = 'employees.json';
    a.click();
    URL.revokeObjectURL(url);
});
```

style.css

```
* { margin: 0; padding: 0; box-sizing: border-box; }
body {
    font-family: 'Poppins', sans-serif;
    background: linear-gradient(135deg, #0f0c29, #302b63, #24243e);
    color: #e8e8f5;
    min-height: 100vh;
    padding: 40px 20px;
}
.container { max-width: 1250px; margin: 0 auto; }
header { text-align: center; margin-bottom: 40px; }
header h1 {
    font-size: 3rem;
    background: linear-gradient(90deg, #4facfe, #00f2fe);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}
header p { color: #b8b8d8; font-size: 1.1rem; margin-top: 8px; }
.grid { display: grid; grid-template-columns: 1fr 2fr; gap: 30px; align-items: start; }
.card {
    background: rgba(255, 255, 255, 0.07);
    border: 1px solid rgba(255, 255, 255, 0.12);
    border-radius: 24px;
    padding: 32px;
    backdrop-filter: blur(12px);
}
.card h2 { font-size: 1.5rem; margin-bottom: 24px; }
.input-group { margin-bottom: 20px; }
.input-group label { display: block; margin-bottom: 8px; font-size: 0.95rem; color: #d0d0ea; }
.input-group input {
    width: 100%;
    padding: 18px 20px;
    border-radius: 14px;
    border: 1px solid rgba(255, 255, 255, 0.15);
    background: rgba(15, 12, 41, 0.7);
    color: #fff;
    font-family: inherit;
    font-size: 1rem;
}
.input-group input::placeholder { color: #8888aa; }
.btn {
    border: none;
    border-radius: 14px;
    padding: 14px 22px;
    font-family: inherit;
    font-size: 1rem;
```

```
    font-weight: 600;
    color: #fff;
    cursor: pointer;
    transition: transform 0.2s;
}
.btn:hover { transform: translateY(-2px); }
.btn-green { width: 100%; padding: 20px; background: linear-gradient(90deg, #11998e, #38ef7d); }
.btn-blue { background: linear-gradient(90deg, #2563eb, #4f46e5); }
.btn-purple { background: linear-gradient(90deg, #7c3aed, #c026d3); }
.table-header, .json-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 20px;
}
table { width: 100%; border-collapse: collapse; }
th {
    text-align: left;
    padding: 14px 16px;
    color: #4facfe;
    text-transform: uppercase;
    letter-spacing: 1px;
    font-size: 0.85rem;
    border-bottom: 2px solid rgba(255, 255, 255, 0.2);
}
td { padding: 18px 16px; border-bottom: 1px solid rgba(255, 255, 255, 0.1); }
.empty-msg { text-align: center; color: #9a9ac0; padding: 25px 0; }
.json-output { margin-top: 30px; padding-top: 25px; border-top: 1px solid rgba(255, 255, 255, 0.1); }
.json-header h3 { color: #b8b8d8; }
pre {
    background: #1a1a2e;
    color: #4ade80;
    padding: 20px;
    border-radius: 14px;
    font-family: 'Courier New', monospace;
    max-height: 200px;
    overflow: auto;
}
@media (max-width: 900px) { .grid { grid-template-columns: 1fr; } }
```

employees.json

```
[
  { "id": "101", "name": "Alice Johnson", "department": "Engineering", "salary": "75000" },
  { "id": "102", "name": "Bob Smith", "department": "Marketing", "salary": "62000" },
  { "id": "103", "name": "Clara Davis", "department": "Human Resources", "salary": "58000" },
  { "id": "104", "name": "David Lee", "department": "Finance", "salary": "80000" },
  { "id": "105", "name": "Eva Martinez", "department": "Engineering", "salary": "90000" }
]
```

Part 2 — Online Food Order Dashboard (jQuery)

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Food Order Dashboard</title>
  <!-- jQuery Library -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      min-height: 100vh;
      padding: 30px;
    }
    .container {
      max-width: 1200px;
      margin: 0 auto;
      background: white;
      border-radius: 15px;
```

```
  box-shadow: 0 10px 40px rgba(0, 0, 0, 0.2);
  padding: 40px;
}
h1 {
  text-align: center;
  color: #333;
  margin-bottom: 10px;
  font-size: 2.5em;
}
h2 {
  text-align: center;
  color: #666;
  margin-bottom: 30px;
  font-weight: normal;
}
.btn-container {
  text-align: center;
  margin-bottom: 30px;
}
button {
  padding: 12px 30px;
  font-size: 18px;
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  color: white;
  border: none;
  border-radius: 25px;
  cursor: pointer;
  transition: transform 0.2s, box-shadow 0.2s;
  box-shadow: 0 4px 15px rgba(102, 126, 234, 0.4);
}
button:hover {
  transform: translateY(-2px);
  box-shadow: 0 6px 20px rgba(102, 126, 234, 0.6);
}
button:active {
  transform: translateY(0);
}
#orderTable {
  border-collapse: collapse;
  width: 100%;
  margin-top: 20px;
  box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
}
th, td {
  border: 1px solid #ddd;
  padding: 15px;
  text-align: center;
}
th {
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  color: white;
  font-weight: bold;
  text-transform: uppercase;
  letter-spacing: 1px;
}
tr:nth-child(even) {
  background-color: #f8f9fa;
}
tr:hover {
  background-color: #e9ecef;
}
.status-preparing {
  background: #ffc107;
  color: #333;
  padding: 5px 15px;
  border-radius: 20px;
  font-weight: bold;
  display: inline-block;
}
.status-ready {
  background: #28a745;
  color: white;
  padding: 5px 15px;
  border-radius: 20px;
  font-weight: bold;
  display: inline-block;
}
```

```
    .status-delivered {
      background: #17a2b8;
      color: white;
      padding: 5px 15px;
      border-radius: 20px;
      font-weight: bold;
      display: inline-block;
    }
    .summary {
      margin-top: 30px;
      padding: 20px;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
      border-radius: 10px;
      text-align: center;
    }
    .summary h3 {
      color: #333;
      margin-bottom: 15px;
    }
    .summary p {
      font-size: 18px;
      color: #555;
      margin: 10px 0;
    }
    .summary .highlight {
      font-weight: bold;
      color: #667eea;
      font-size: 22px;
    }
    #errorMessage {
      display: none;
      text-align: center;
      color: #dc3545;
      font-size: 18px;
      padding: 20px;
      background: #f8d7da;
      border-radius: 10px;
      margin-top: 20px;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>&#127869;&#65039; Online Food Order Dashboard</h1>
    <h2>Real-time Order Management</h2>

    <div class="btn-container">
      <button id="loadBtn">View Orders</button>
    </div>

    <table id="orderTable">
      <thead>
        <tr>
          <th>Order ID</th>
          <th>Customer Name</th>
          <th>Food Item</th>
          <th>Quantity</th>
          <th>Total Amount</th>
          <th>Order Status</th>
        </tr>
      </thead>
      <tbody id="tableBody">
      </tbody>
    </table>

    <div id="errorMessage">
      <strong>&#9888;&#65039; Error:</strong> Unable to load orders.json. Please try again later.
    </div>

    <div class="summary" id="summary" style="display: none;">
      <h3>Order Summary</h3>
      <p>Total Number of Orders: <span class="highlight" id="totalOrders">0</span></p>
      <p>Total Sales Amount: <span class="highlight" id="totalSales">$0.00</span></p>
    </div>
  </div>

  <script>
```

```
    $("#loadBtn").click(function() {
      $.getJSON("orders.json", function(data) {
        $("#tableBody").empty();
        var totalOrders = 0;
        var totalSales = 0;

        $.each(data, function(index, order) {
          var statusClass = "";
          if (order.status === "Preparing") {
            statusClass = "status-preparing";
          } else if (order.status === "Ready") {
            statusClass = "status-ready";
          } else if (order.status === "Delivered") {
            statusClass = "status-delivered";
          }

          $("#tableBody").append(`
            <tr>
              <td>${order.orderId}</td>
              <td>${order.customerName}</td>
              <td>${order.foodItem}</td>
              <td>${order.quantity}</td>
              <td>$${order.totalAmount.toFixed(2)}</td>
              <td><span class="${statusClass}">${order.status}</span></td>
            </tr>
          `);
          totalOrders++;
          totalSales += order.totalAmount;
        });

        $("#totalOrders").text(totalOrders);
        $("#totalSales").text("$" + totalSales.toFixed(2));
        $("#summary").show();

      }).fail(function() {
        $("#errorMessage").show();
        $("#summary").hide();
      });
    });
  </script>
</body>
</html>
```

orders.json

```
[
  { "orderId": "ORD001", "customerName": "Aarav Sharma", "foodItem": "Margherita Pizza", "quantity": 2, "totalAmount":
  { "orderId": "ORD002", "customerName": "Priya Patel", "foodItem": "Veg Burger", "quantity": 1, "totalAmount": 8.49,
  { "orderId": "ORD003", "customerName": "Rohan Mehta", "foodItem": "Pasta Alfredo", "quantity": 3, "totalAmount": 35.
  { "orderId": "ORD004", "customerName": "Sneha Joshi", "foodItem": "Fried Rice", "quantity": 2, "totalAmount": 18.50,
  { "orderId": "ORD005", "customerName": "Karan Desai", "foodItem": "Paneer Tikka", "quantity": 4, "totalAmount": 42.0
  { "orderId": "ORD006", "customerName": "Meera Nair", "foodItem": "Chicken Biryani", "quantity": 1, "totalAmount": 14
  { "orderId": "ORD007", "customerName": "Arjun Rao", "foodItem": "Masala Dosa", "quantity": 2, "totalAmount": 12.00,
]
```

8. Output:

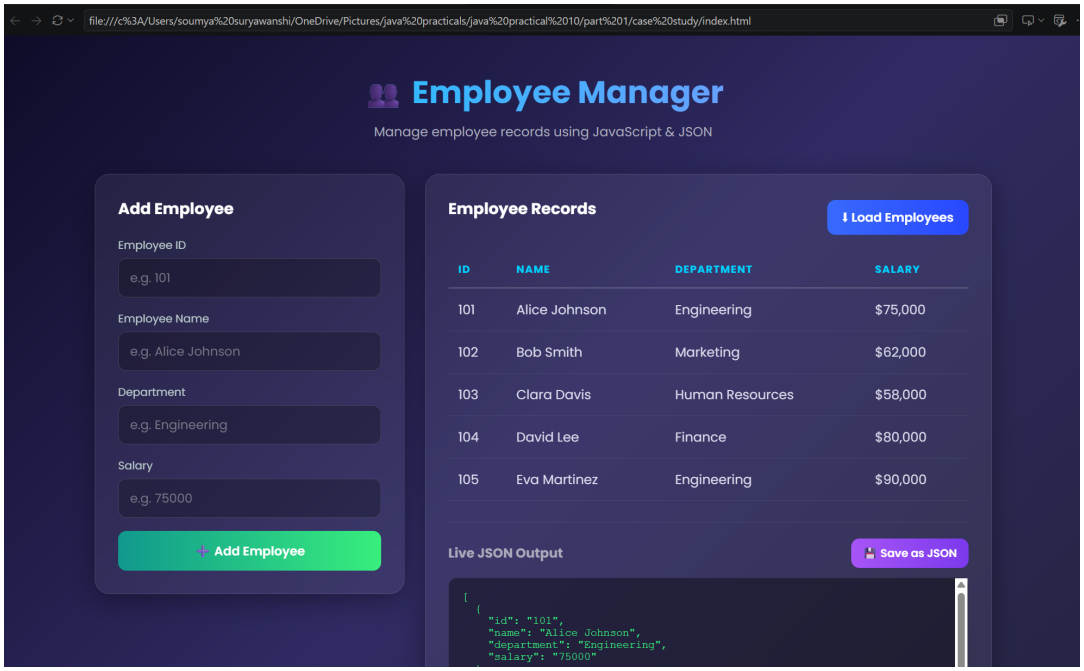


Fig 3: Part 1 case study (Employee Manager) output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 10/part 1/case study/index.html

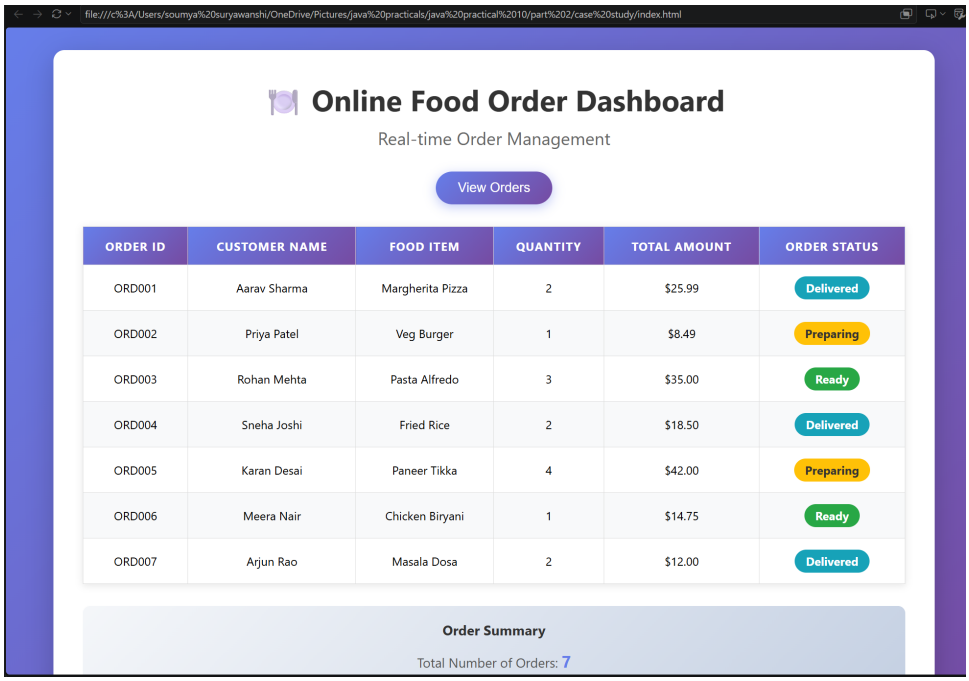


Fig 4: Part 2 case study (Online Food Order Dashboard) output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 10/part 2/case study/index.html

9. Result/Conclusion:

The experiment successfully demonstrates loading external JSON data into a web page without reloading it. In Part 1, the Fetch API was used to request `students.json`, parse the response with `response.json()` and build a student table dynamically. In Part 2, the same task was completed with jQuery's `$.getJSON()`, `$.each()` and `.append()`, which required less code. In the accompanying case studies, an Employee Manager was built that adds records through a form, shows them as live JSON with `JSON.stringify()`, loads `employees.json` using Fetch and saves the data as a JSON file, and an Online Food Order Dashboard was built that loads `orders.json` with jQuery, shows colour-coded order statuses and calculates the total number of orders and total sales. Both implementations produced the expected output for the given inputs and reinforced practical understanding of JSON handling, asynchronous requests and DOM manipulation using Fetch and jQuery.

Upload the Soft copy as well as complete code on GitHub link is mandatory.